Business Insights and Recommendations Report

CS Vision – Data Analytics Internship Program (Set A)

Project: Financial & Operational Performance Analytics System

Week 3: KPI Development and Profitability Analysis (3 – 9 November 2026)

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1. Objective

This report develops measurable business KPIs from the cleaned Global Superstore dataset (51,290 order lines, 2011–2014), analyses profitability by product, market, region and discount level, calculates growth rates, detects anomalies, and presents ten evidence-based insights with practical recommendations. The notebook is *Week3_KPI_Profitability.ipynb*; charts 16–23 are in the visualizations folder.

2. Key Performance Indicators

Definitions: Cost = Sales - Profit; Gross Profit = Profit; Net Profit = Profit - Shipping Cost; Total Expenses = Cost + Shipping Cost.

KPI	Value	Meaning
Total Revenue	12,642,502	Sum of Sales
Total Expenses	12,527,865	Cost 11,175,045 + Shipping 1,352,821
Gross Profit	1,467,457	Profit before shipping
Net Profit	114,637	After shipping expense
Gross Margin	11.61%	Gross Profit / Revenue
Net Profit Margin	0.91%	Net Profit / Revenue
Revenue Growth Rate (CAGR 2011–2014)	23.92%	Compound yearly growth
Average Transaction (Order) Value	504.99	Revenue / 25,035 orders
Cost-to-Revenue Ratio	88.39%	Cost / Revenue (99.09% including shipping)
Loss-making Order Lines	41.24%	Lines with negative net profit

Product and regional contribution are reported in Sections 5 and 6. The KPI table is also saved as *kpi_summary.csv* for the Week 4 dashboard.

3. Growth Analysis

Year	Revenue	Growth	Net profit	Gross margin	Net margin
2011	2,259,451	–	4,670	11.0%	0.21%
2012	2,677,439	+18.5%	23,923	11.5%	0.89%
2013	3,405,746	+27.2%	42,385	11.9%	1.24%
2014	4,299,866	+26.3%	43,658	11.7%	1.02%

Revenue has almost doubled, but net margin peaked in 2013 and fell in 2014. Month-over-month growth averages 10.2% but is volatile: only 26 of 47 months grew, with the largest rise in August 2012 (+108.7%) and the largest fall in January 2012 (-59.3%).

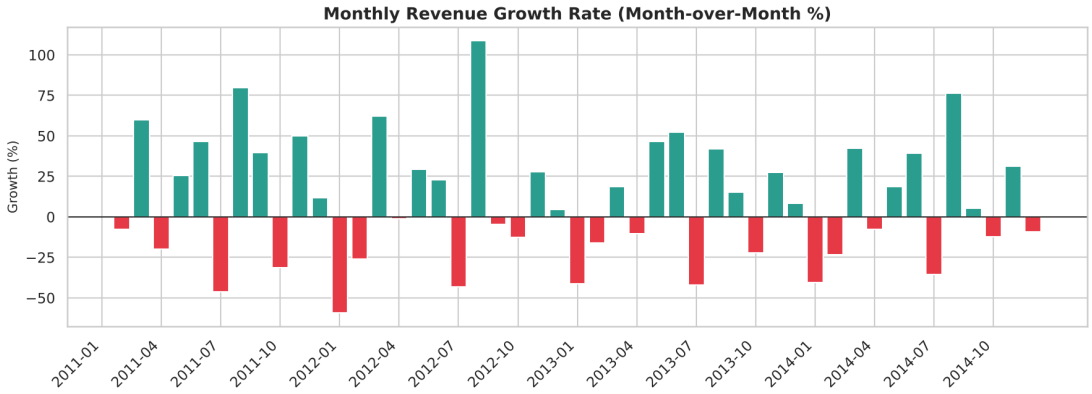


Chart 16: Monthly revenue growth rate (month-over-month)

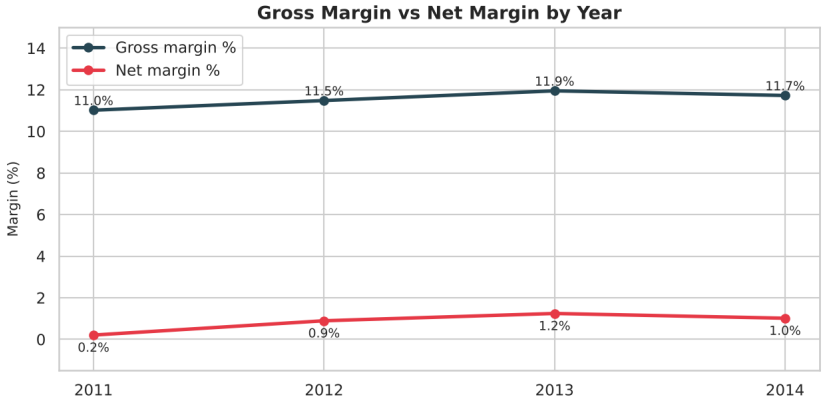


Chart 17: Gross margin vs net margin by year

4. Profitability Analysis

Of every 100 units of revenue, about 88 go to cost, leaving a gross margin of 11.6%. Shipping then takes 10.7, so the business keeps under 1 unit.

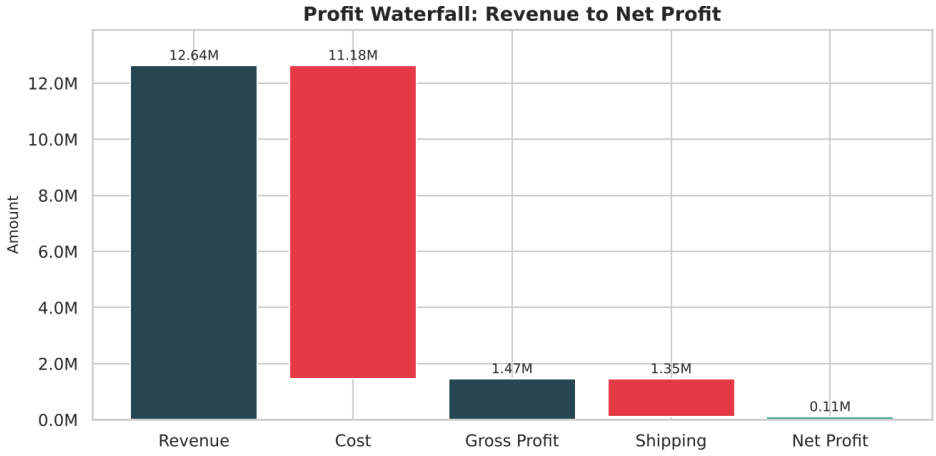


Chart 18: From revenue to net profit

5. Product-Level Profitability

Of 3,788 products, 2,351 are profitable and 1,437 (37.9%) lose money. The top 10 products represent only 4.9% of sales, so the catalogue is highly fragmented. The Canon imageCLASS 2200 Advanced Copier earns the most (23.9K, 38.8% margin), while the Cubify CubeX 3D Printer Double Head Print loses the most (-9.6K, -86% margin).

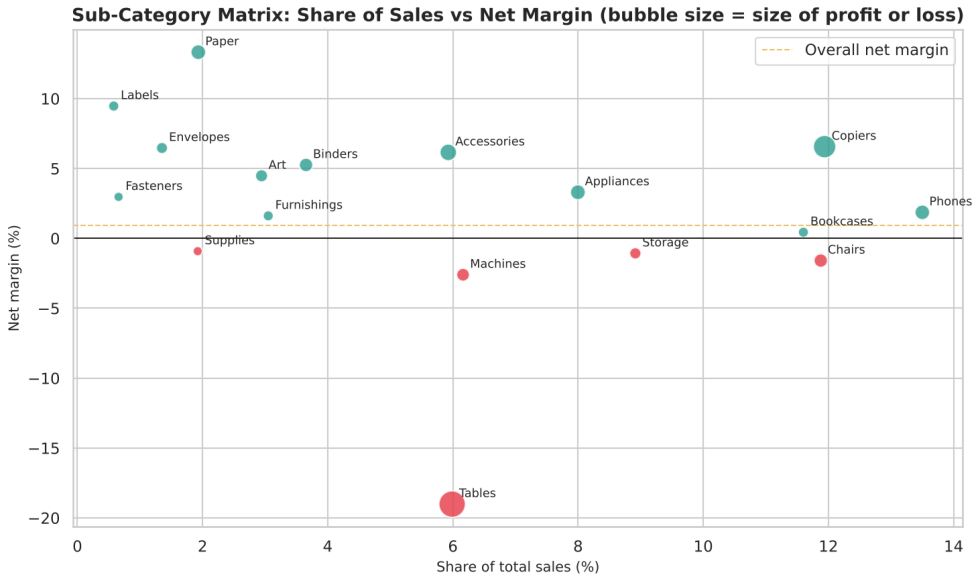


Chart 19: Sub-category matrix – share of sales vs net margin

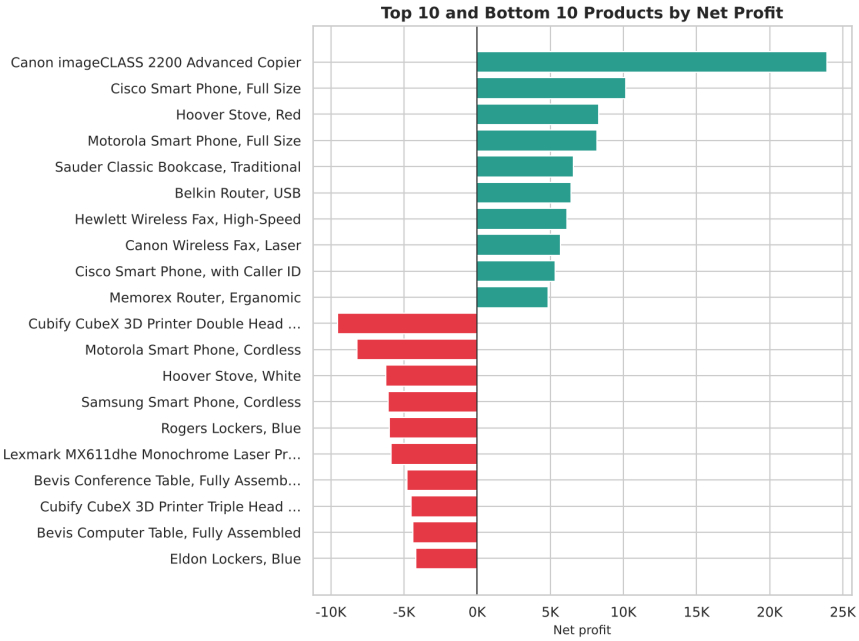


Chart 23: Top 10 and bottom 10 products by net profit

6. Market and Regional Profitability

Market	Share of sales	Net profit	Net margin
APAC	28.4%	48,835	1.4%
EU	23.2%	63,407	2.2%
US	18.2%	48,223	2.1%
LATAM	17.1%	-12,495	-0.6%
EMEA	6.4%	-44,478	-5.5%
Africa	6.2%	732	0.1%
Canada	0.5%	10,412	15.6%

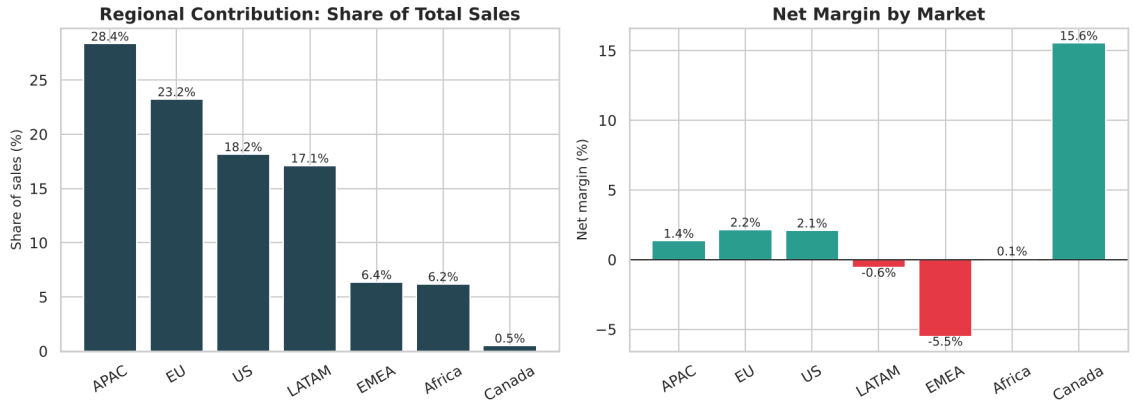


Chart 20: Regional contribution and net margin by market

7. Discount Impact

Lines discounted above 20% are 22.1% of all lines and 15.3% of sales, but they have a net margin of -52.8% and lose 1,018,666 in total. The remaining lines earn a 10.6% margin and 1,133,303 of profit. The heatmap shows that every category turns loss-making above a 20% discount.

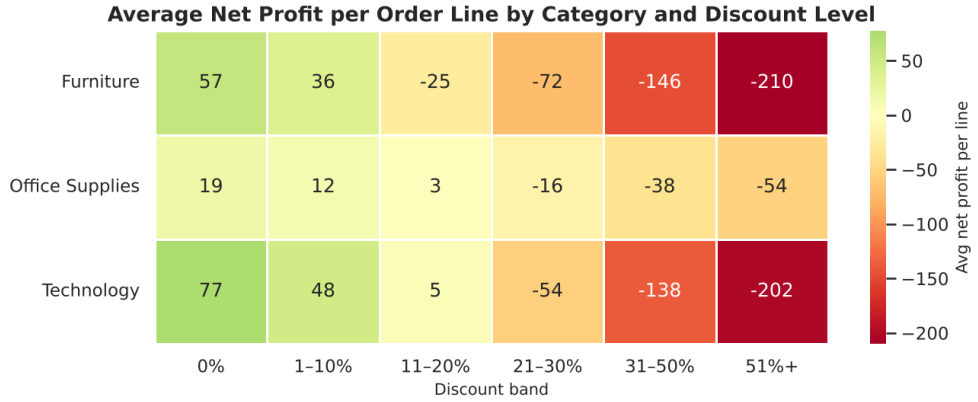


Chart 21: Average net profit per order line by category and discount level

8. Anomaly Detection

Two methods were used. The z-score method (more than 3 standard deviations from the mean) flags 1,435 lines (2.8%): 820 on net profit and 1,020 on sales. The IQR rule within each sub-category flags 1,605 sales and 3,106 profit outliers. The z-score anomalies have a combined net profit of -19,778, while all other lines earn +134,414.

- The 10 largest losses (from -2.9K to -7.1K) all had discounts of 50–80%, and most are 3D printers, binding systems, smart phones and large printers.
- 2,104 lines with discounts of 70% or more lose 245,860 in total.
- 15 lines have shipping costs above 50% of the sale value.
- September and October 2014 had unusually high profit; December 2014 had an unusual loss (-7.9K) despite 503K in sales.

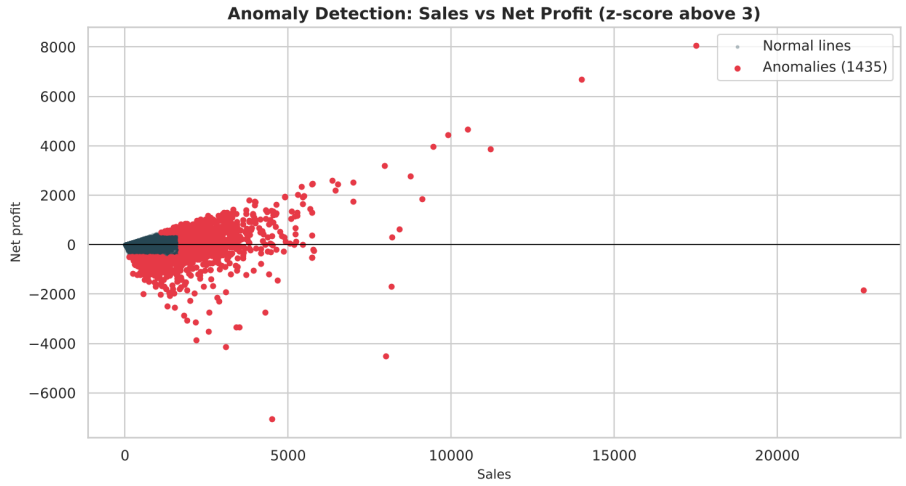


Chart 22: Anomalies – sales vs net profit (z-score above 3)

9. Business Insights

#	Insight and evidence
1	**Strong, order-driven growth.** Revenue grew from 2.26M to 4.30M (CAGR 23.9%), but average order value stayed flat at about 505.
2	**Margins are very thin.** Gross margin is 11.6% and net margin 0.91%; expenses absorb 99.1% of revenue. Net margin fell from 1.24% (2013) to 1.02% (2014) while revenue grew 26%.
3	**Discounts above 20% are the biggest profit leak.** 22.1% of lines, margin -52.8%, total loss 1.02M; the other lines earn 10.6%.
4	**Four sub-categories lose money.** Tables (-143.9K, -19%), Chairs (-23.8K), Machines (-20.3K) and Storage (-12.1K) lose about 200K together.
5	**Profit is concentrated in Copiers.** Copiers earn 99.1K (6.6% margin), 86% of total net profit.
6	**A long tail of loss-making products.** 37.9% of products lose money; the top 10 are only 4.9% of sales.
7	**Markets differ sharply.** EU and US earn about 2%; LATAM (-0.6%) and EMEA (-5.5%) lose money; APAC is the largest market at only 1.4%.
8	**Peak season does not convert to profit.** November 2014 had 555K in sales but only 2.9K profit; December 2014 lost 7.9K.
9	**Anomalies are mostly extreme discounts.** Lines with discounts of 70% or more lose 245.9K; the 10 largest losses all had 50–80% discounts.
10	**Shipping is a major cost.** It is 10.7% of sales, and 41.2% of lines lose money after shipping; Same Day and First Class cost about 17% of sales against 8% for Standard Class.

10. Recommendations

Priority	Recommendation	Evidence	KPI to track
1	Cap discounts at 20% and require manager approval above it.	Insights 3, 9: up to 1.02M of losses, about 9 times current net profit.	Share of lines above 20% discount; net margin
2	Reprice or restrict discounts on Tables, Chairs, Machines and Storage.	Insight 4: about 200K of losses.	Sub-category net margin
3	Introduce shipping surcharges or free-shipping thresholds, and steer customers to Standard Class.	Insight 10: shipping is 10.7% of sales.	Shipping cost % of sales
4	Review pricing and shipping in LATAM and EMEA; improve APAC margin.	Insight 7.	Net margin by market
5	Review and prune products with persistent losses; protect Copiers and high-margin technology.	Insights 5, 6.	Number of loss-making products
6	Avoid deep Q4 discounts and plan stock and promotions earlier.	Insight 8.	Q4 net margin
7	Add automatic alerts for discounts of 50% or more and for lines that would sell at a loss.	Insight 9.	Number of anomaly lines
8	Encourage larger baskets with bundles and cross-selling.	Insight 1.	Average order value

11. Assumptions and Limitations

- Cost and Net Profit are derived. If the source Profit already includes shipping, the net margin is understated.
- The recommended impacts are upper bounds based on historical data; real savings depend on customer behaviour.
- The dataset has no Order Status or Payment Status field, and covers 2011–2014 only.
- Anomalies are statistical flags and should be reviewed before being treated as errors; many are genuine large orders.

12. Conclusion

Revenue is growing quickly, but profit is not: discounting above 20%, shipping costs and a small number of loss-making sub-categories and markets absorb almost the entire gross margin. Acting on the first three recommendations would have the largest effect on profitability. These KPIs will form the cards and filters of the Week 4 dashboard.